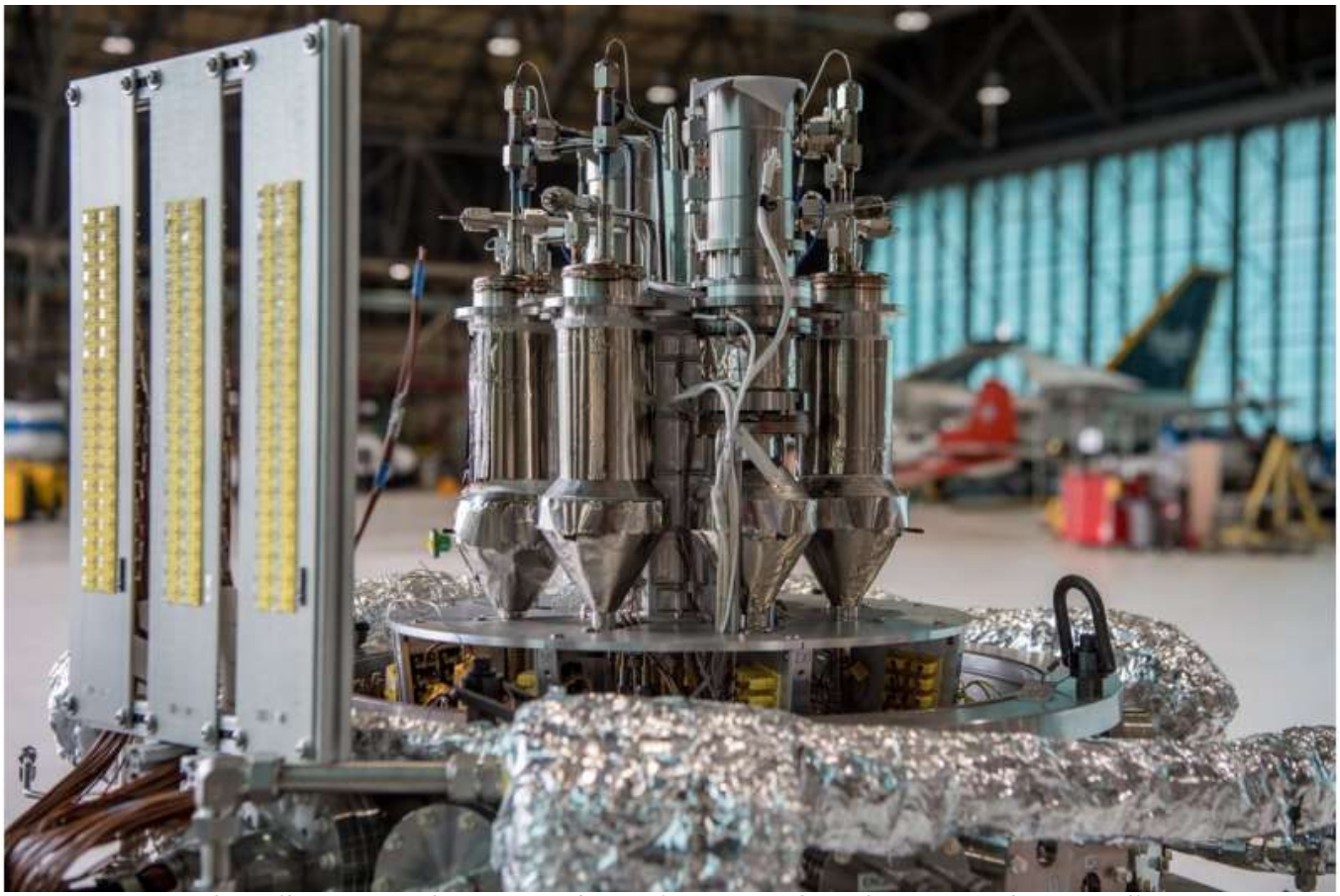
Deepening Stirling engine analysis: Optimized model offers more accurate performance predictions

Matthew Williams



Prototype NASA 1kW Kilopower nuclear reactor for use in space and on planetary surfaces. Credit: NASA Glenn Research Center

Multiple space agencies plan to return astronauts to the moon by the end of this decade. Along with commercial and international partners, these efforts aim to create infrastructure that will ensure a "sustained program of lunar exploration and development." This includes NASA's Artemis Program, China's International Lunar Research Station (ILRS), and the ESA's Moon Village, all of which consist of creating lunar habitats around the South Pole-Aitken Basin. Providing power for these bases is a significant challenge given the cycle of lunar day and night, which lasts for two weeks at a time.

Several promising technologies are being investigated to address this challenge, including Stirling engines, which could power space nuclear reactor power systems (SNRPS). However, many properties and design considerations must be considered before functional prototypes are built.

In a recent paper published in *Nuclear Science and Techniques*, a team of Chinese scientists created an analytical model to evaluate different Stirling engine designs and determine which is the most promising. Their work presents a Stirling cycle analysis method that more accurately captures the engine's real-world operating behavior.

The study was led by Shang-Dong Yang, a professor of organic chemistry with the College of Nuclear Technology and Automation Engineering (CNTAE) at the Chengdu University of Technology. He was joined by fellow researchers from the CNTAE and the Science and Technology on Reactor System Design Technology Laboratory at the Nuclear Power Institute of China.

A Stirling engine is a closed-cycle regenerative heat system that utilizes the expansion and contraction of gases

(exposed to different temperatures) to convert heat energy into mechanical work. These engines are known for their high efficiency and versatility, making them prime candidates for advanced power systems in extraterrestrial environments. Unfortunately, predicting their potential performance in environments like the moon and Mars remains difficult since real-world testing data is lacking.

In the meantime, scientists are forced to rely on theoretical models that take into account the thermodynamics of such a system. In this case, second-order analysis methods are used extensively to inform the design and thermodynamic analysis of Stirling engines. For their study, the team developed a simplified version that bridges thermodynamic cycles and engine operation by accounting for various energy dissipation factors, including shuttle heat loss, seal leakage, flow resistance, and finite piston speed.

These considerations are crucial when designing engines that can function and provide power to lunar and Martian habitats and other facilities necessary for working and living beyond Earth. As Prof. Fong explained in a EurekaAlert news release: "Our refined model offers a clearer picture of how various design parameters, such as regenerator porosity and working fluid choice, affect Stirling engine efficiency and power output. This advancement provides critical reference and data support for the application of Stirling engines in advanced compact energy systems."

The predictive capability of their model is also validated based on experimental data from existing Stirling engines, like the GPU-3 and the free-piston RE-1000. NASA developed these concepts during the 1970s to produce applications for future missions that would build on the accomplishments of the Apollo Program.

The next step for the team is to leverage this model to explore dynamic operational scenarios like engine start-up and transient responses. This research could lead to prototypes generating electricity for compact nuclear reactors, which could be tested in simulated lunar and Martian conditions. With luck, the technology could assist in the creation of permanent human outposts on the moon and provide applications for life here on Earth.

Said Prof. Fong, "Future research will focus on understanding and managing thermal balance across all operational stages within an integrated reactor system, with particular attention to start-up in sensitive environments such as space. This includes investigating the influence of the heat pipe reactor's output characteristics on the Stirling engine's performance, efficiency, and stability."

More information: Shang-Dong Yang et al, Study on the operating characteristics of Stirling engine based on an optimized analysis method, *Nuclear Science and Techniques* (2025). DOI: 10.1007/s41365-025-01711-6

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